FROM MASSES TO AGES

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ACKNOWLEDGMENT

ABSTRACT

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CHAPTER 1

Introduction

There is currently no way to systematically determine the age of a planet. In the field of planet formation, the standard practice is assuming coevality: that is, the planet and its host star are the same age. However, we depend on the age of the planet, along with the luminosity that can be measured from direct imaging, to estimate the mass using evolutionary models. However, it is obvious that planet formation takes time, possibly millions of years, so it follows that a lot of mass estimates we obtain are based upon incorrect ages. This project aims to tackle this research gap by finding a systematic method to estimate the age of a planet, using dynamical masses, luminosities from direct imaging and evolutionary models. Before the approach is explored, below is the background required to understand what's required to undertake this project: how planets form and their respective timescales, how planets are imaged, the aforementioned evolutionary models, a deeper dive into the motivation and then finally the methodology.

1.1. Planetary System Formation

Planetary systems are born from a planet-forming disk around a young star, which form from a giant molecular cloud ($\sim 10^6 M_\odot$, $\sim 10^7$ AU) contracting under its own self gravity (Shu et al., 1987). This contraction forms a protostar with a surrounding rotating envelope. Due to the conservation of angular momentum, excess mass from the envelope not accreted by the star will settle into a rotating disk of gas and dust, which is the planet-forming (protoplanetary) disk of interest. This disk has an expected lifetime of ~ 10 Myr (Strom et al., 1993), during which it will form planets. Once exhausted of its material, the disk will dissipate and a planetary system much like our own solar system will result.

1.2. Determination of Stellar Age

The ages of young stars can be determined by a few methods, three of which I will mention in this section as they are of particular relevance to the systems studied in this project.

1.2.1. Lithium Depletion Boundary

This process only happens in groups of stars with the same age, known as moving groups. Young stars contract as they age; in doing so, their core temperatures rise (Soderblom et al., 2014). Lithium burning occurs at about 3×10^6 K, and these young stars reach this temperature on a timescale that depends on mass. We can tell if this has occurred through spectroscopy. The lithium depletion boundary (LDB) is the between low luminosity stars still have all their lithium, and slightly higher luminosity stars have totally burned through their lithium.

1.2.2. Kinematic Ages

1.2.3. Hertzsprung-Russell Diagrams

1.3. Planetary Formation Mechanisms

1.3.1. Core Accretion

The standard model for gas-giant formation by core accretion comes from Pollack et al. (1996), who built on older accretion models (Safronov, 1969; Mizuno et al., 1978; Mizuno, 1980; Bodenheimer and Pollack, 1986). A general overview of the process is as follows: a protoplanetary disk consists of dust and gas. This dust undergoes random motion, and eventually dust particles may collide and clump together. The repetition of this process eventually leads to pebbles (dust coagulation, Drążkowska et al., 2023), then boulders, plentesimals, then finally, through planetesimal accretion (Safronov, 1969), a planetary core massive enough to start accreting gas from the disk, which eventually leads to runaway gas accretion (Mizuno, 1980, Bodenheimer and Pollack, 1986, Pollack et al., 1996).

All in all, this process' timescale is on the scale of Myr (Pollack et al., 1996), which is significant because a planet formed in this way would be sizeably younger than its host star; planets formed

through CA are also what this project aims to find.

This process occurs closer to the centre of the disk than gravitational instability, the other widely-accepted formation pathway, detailed in the next section. Note that this process is not necessarily linear; most pebbles actually do not evolve to the metre-scale. Among the potential barriers include the fragmentation-barrier, drift-barrier and others that are beyond the scope of this project but are detailed in Drazkowska et al. 2023.

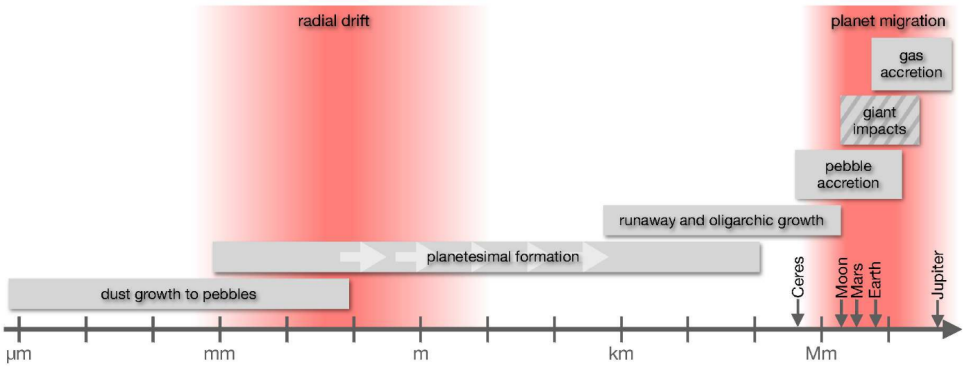


Figure 1.1: Fig. 3 from Drazkowska2023. This figure provides a general scale for the stages of CA.

Previous models that Pollack et al., 1996 model built upon include Mizuno et al., 1978 and Mizuno, 1980 of core instability, Bodenheimer and Pollack, 1986's investigations into Mizuno's models, Pollack et al., 1986 and Podolak et al., 1988 examinations into planetesimal accretion, among others.

In short, a critical mass before runaway gas accretion occurs is found to be $\sim 10M_{\oplus}$ (Mizuno, 1980), later $\sim 10M_{\oplus} - 30M_{\oplus}$ (BP86), and also when the heavy-element mass and the gas mass of the planet would equal; after this critical mass is reached, runaway gas accretion occurs. However, observations of the giant planets also revealed that Neptune and Uranus had compositions such that, in these models, their mass fractions were comprised of too little gas mass, so they would have never reached the equality required to reach the critical mass. Pollack et al., 1996's remedy to this issue will be found in the outline below. Note also, that this model occurs after dust and pebble accretion and

Arranged neatly, the model of Pollack et al., 1996 is split into three phases: during phase 1, planetesimal accretion exceeds gas accretion rate; they're initially many orders of magnitude apart, with planetesimal accretion rate being the greater value. Overtime, the two rates become comparable: after $\sim 10^5$ years, they are comparable and planetesimal accretion rate begins declining sharply. Meanwhile, gas accretion rate keeps growing. This phase denotes a runaway planetesimal accretion, separate from the runaway gas accretion, and it finishes when a protoplanet has emptied its feeding zone of planetesimals. However, the feeding zone grows as the planet's total mass increases, introducing new planetesimals (Pollack et al., 1996)

Phase 2 is a novel phase introduced in this model, that denotes a long, liminal phase where both accretion rates are relatively time-invariant, now with gas accretion rate exceeding planetesimal accretion rate. This phase spans the bulk of the aforementioned Myr timescale, with Pollack's simulations of Jupiter lasting $\sim 7Myr$. Finally, phase 3 denotes the runaway gas accretion, where both accretion rates (with gas still far exceeding planetesimal) start rapidly increasing again.

Fig 1c of Pollack et al., 1996 is particluarly relevant to this project: during phase 3, the luminosity grows rapidly such that, once fully formed, the planet is bright enough that we can directly image it. It will then cool down for the rest of its lifetime (since planets have no mechanisms to self-illuminate).

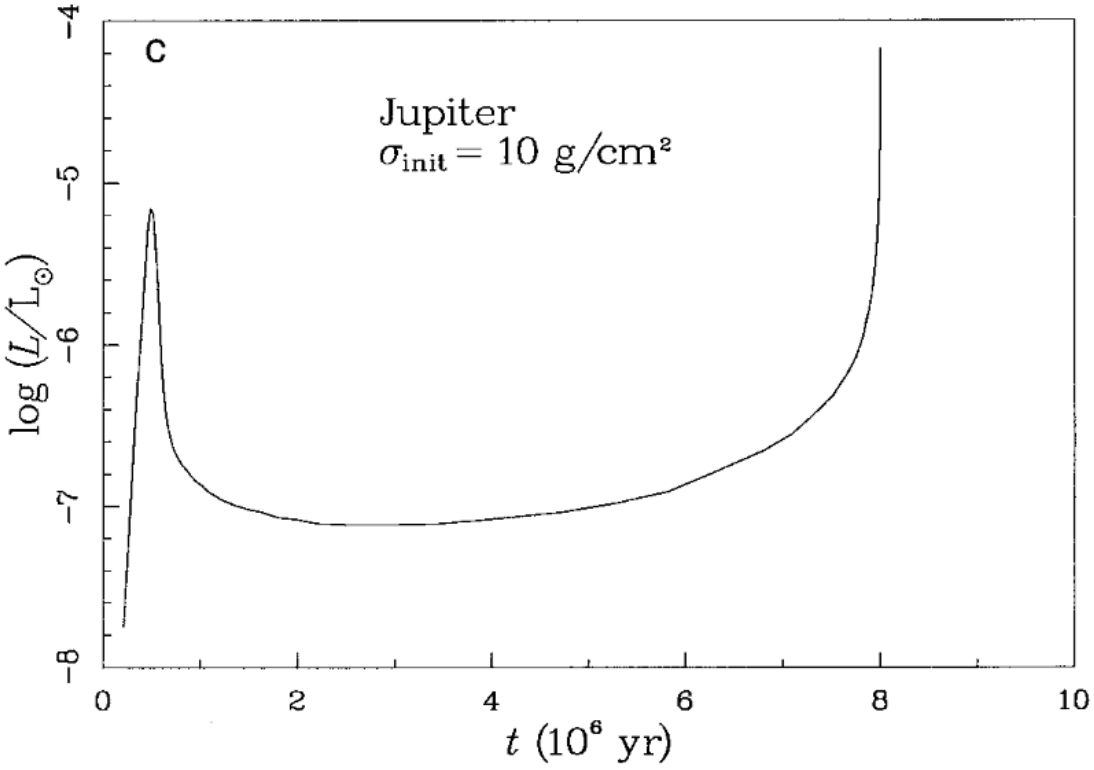


Figure 1.2: Fig 1c of Pollack et al., 1996. The initial rise corresponds to the runaway planetesimal accretion of phase 1; the long plateau corresponds to the transitionary phase 2; the sharp increasing corresponds to both accretion rates rapidly increasing (and the runaway gas accretion) in phase 3

1.3.2. Gravitational Instability

Planet formation by gravitational (disk) instability arises in the outer regions of the disk (Dodson-Robinson et al., 2009) where it is colder and results from a clump fragmenting from the disk under its own self-gravity. Its dependence on being cold is due to the Toomre stability criterion Q, given by:

$$Q = \frac{c_s \Omega}{\pi G \Sigma}$$

where c_s is the speed of sound, Ω is rotation frequency, Σ is the disk surface density (Toomre, 1964; Kratter and Lodato, 2016; Longarini and Lodato, 2026). Q determines the stability of a disk: large Q is stable, and $Q = 1$ is the instability criterion: any $Q < 1$ is unstable. The speed of sound scales with temperature, so Q scales linearly with sound speed and thus temperature, and inversely with

surface density, so cold massive disks are more prone to instability. The parameters of Q are in a three-way competition: Σ tries to collapse the clump, c_s tries to resist collapse on small scales, Ω resists on large scales.

So, instability depends on surface density, rotation and temperature that can create denser regions in the disk, that can then fragment into its own subcloud, which will be contracting. This may repeat many times for this fragment, forming sub-subclouds, then sub-sub-sub... clouds (Whitworth et al., 2006). These sub-sub-sub... clouds will contract continually over a few thousand years to a million years, scaling inversely with mass (so more massive clouds contract faster). Here, the subcloud is made of H_2 and is $\sim 10^2$ R_{Jup} in size, and will continue to contract, increasing in temperature in the process until a critical temperature of $\sim$ 2000 K (Helled et al., 2014; Bussmann and Helled, 2026) is reached, where the H_2 dissipates and the now protoplanet undergoes a dynamical collapse, rapidly contracting into a dense, compact gas giant of a few R_{Jup}. It will then cool in a similar fashion to core accretion.

When observing protoplanetary disks, an indicator of disk instability occurring is spiral arms, characterised by transporting angular momentum throughout the disk, and can be driven by increasing surface density or decreasing temperature. These formation mechanisms can be seen in Figure 1.3.

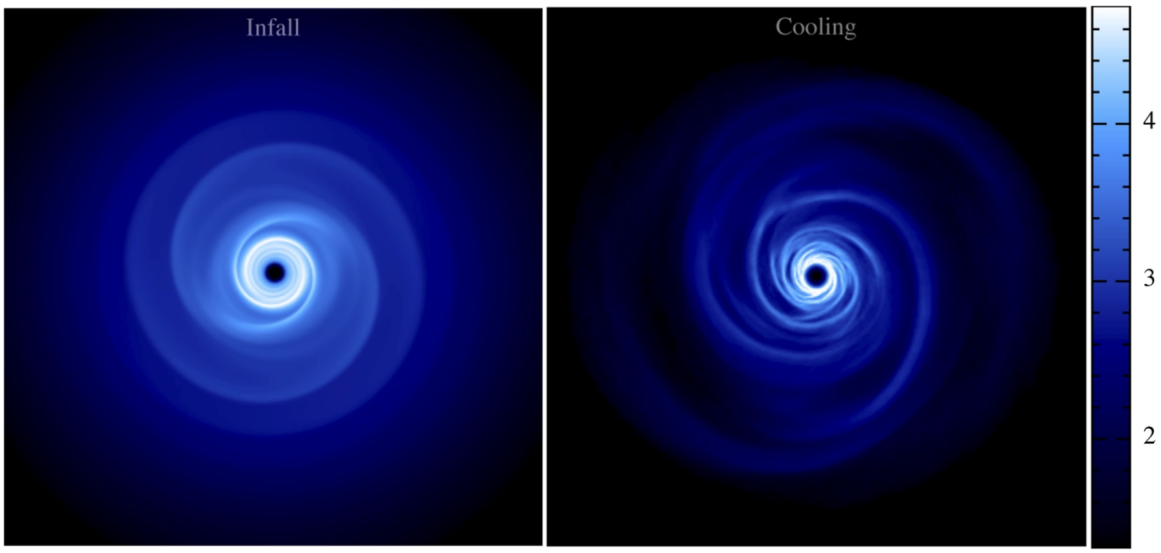


Figure 1.3: Fig. 2 from Longarini & Lodato 2026. It shows two potential transport mechanisms that drive the formation of spiral arms, where infall is from increasing surface density (Σ) or cooling (decreasing sound speed c_s)

1.3.3. A Note on Cloud Fragmentation and Brown Dwarfs

Cloud fragmentation is traditionally a mechanism that forms stars. A general overview (see Whitworth et al., 2006 for the full process) is that given a molecular cloud is massive enough, it contracts and fragments into subclouds. This process repeats for the subclouds, until eventually the subclouds are so optically thick that they can't contract anymore since they can't radiate away the contraction heat (Whitworth et al., 2006).

The floor of the mass of objects formed seems to reach well into the planetary-mass range ($\sim$ 1 to 4 M_{Jup}, Whitworth and Stamatellos, 2006). Also traditionally, the criterion for brown dwarfs is that it exceeds the deterium burning limit of $\simeq 13 M_{Jup}$ (Chabrier and Baraffe, 2000), however even this remains challenged; Chabrier et al., 2014 argues plantary and brown dwarf can overlap and that there is no clear cut-off mass for one body to be a brown dwarf and another a planet.

Actually, the criterion for what a planet actually is even under debate: some sources base it on only their mass, while some go off the formation tracks. Furthermore, the formation tracks for brown dwarfs are also contested: many studies theorise that brown dwarfs can form from the low-end masses

of cloud fragmentation (Balmer et al., 2024; Chabrier, 2003). Some others suggest they could also be formed through disk instability (Boss, 1997; Stamatellos et al., 2007; Stamatellos and Whitworth, 2009) or even core accretion along with the associated runaway gas accretion (Balmer et al., 2024; D'Angelo et al., 2010; Mollière and Mordasini, 2012).

Thus, this project will also make use of lower mass brown dwarfs under the assumption that they can use the same formation tracks as planets.

1.4. Imaging

1.4.1. Targets

Most of the planetary systems imaged in this project use the Very Large Telescope (VLT), and the Spectro-Polarimetric High-contrast Exoplanet REsearch (SPHERE) instrument, at the European Southern Observatory. SPHERE contains the instruments required to carry out the reduction and imaging techniques required to actually observe the planet, such as the coronagraph and wavefront correctors (Currie et al., 2023). Coronagraphs block out the starlight via physical occulters in the image plane, and deformable mirrors correct for wavefron errors from atmospheric distortion.

We target young, bright, giant planets at wide orbits because they are luminous enough to actually be imaged, and far enough from their host star that starlight doesn't obscure the planet and it is detectable within our telescope resolution limits. Binary systems are ignored (Currie et al., 2023): they can inhibit planetary formation and disturb planetary orbits due to their chaotic nature. In general, the target planets are at separations of 30–300 AU, due to contrast limits at small separations and limited field of view at larger separations.

1.4.2. Dynamical Masses

From imaging, we require the luminosities and dynamical masses from the planets. Dynamical masses are found by fitting the orbits to Kepler's laws through Python packages, e.g., orbitize! (Blunt et al., 2024); the separation distance and rotation angle from North are required from the orbits.

1.5. Evolutionary Models

Evolutionary models are split into three categories: hot-start, warm-start and cold-start models. Hot-start model planets start with high entropy retain high entropy after formation, cold-start models have low entropy, and warm-start models are intermediate between the two former (Bowler, 2016; Bussmann and Helled, 2026). In the physical world, this translates to core accretion being able to be hot-start or cold-start, depending on how much gravitational energy is radiated away during formation, and gravitational instability heavily faovuring hot-start due to there being no known mechanism that removes entropy during collapse (Galvagni et al., 2012; Bussmann and Helled, 2026). This project will make use of hot-start models in the implementation, due to core accretion not being a favoured formation mechanism at the studied planets' separations.

CHAPTER 2

Motivation

One of the major caveats to our current models are that planets and their parent stars are not exactly coeval (Bowler, 2016). Planets can span anywhere from the full stellar age, to 10 Myr younger, depending on the formation mechanism; top-down mechanisms would be close to coeval, while bottom-up mechanisms would leave a discrepancy of a few Myr. Furthermore, depending on the formation mechanism, it could lead to different inferred masses: take for example the young, planetary-mass companion 2M1207–3932 b of the TW Hydrae Association. Assuming coevality, its hot-start mass is $\sim 5M_{Jup}$. But, if the formation is delayed by 8 Myr, then the mass is only $\sim 2.5M_{Jup}$ (Bowler, 2016). Thus,

CHAPTER 3

Literature Review

CHAPTER 4

Methodology

This project makes use of the species Python package (Stolker et al., 2020). The idea is to develop code to brute force an inversion of the evolutionary models, where instead of getting a mass estimate from age and luminosity, we get an age estimate from the luminosity and dynamical mass. The idea is to create "standard" curves, using an evolutionary model (one of ATMO, COND, DUSTY), sampling within the uncertainties of the bolometric luminosity of the planet via Monte Carlo sampling and a pre-defined array of ages. The curves will be different for every planet but the implementation is the same.

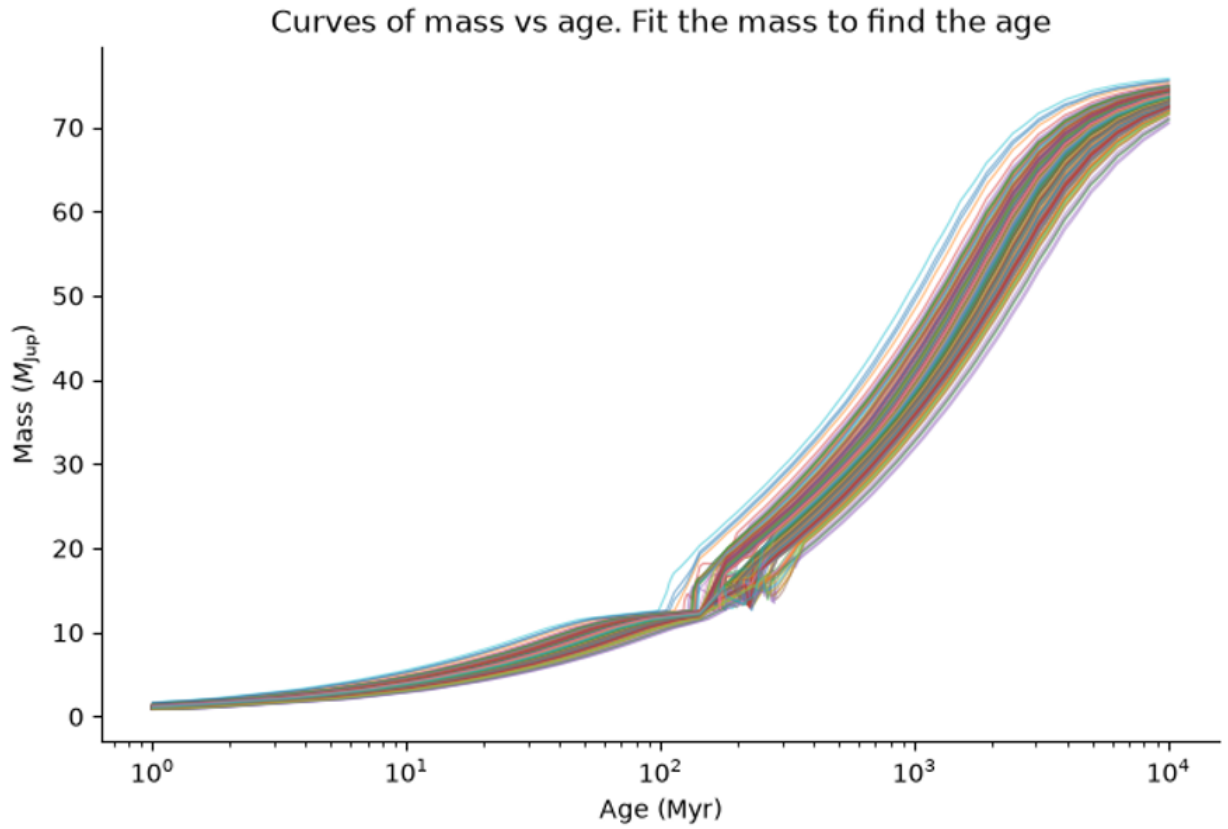


Figure 4.1: Standard curves for AF Lep b. Generated with $\log \frac{L}{L_\odot} = -4.81 \pm 0.13$ and a logspace of ages from 0 to 10 Gyr.

The same Monte Carlo sampling is applied to the dynamical mass and its respective uncertainties, which is fitted to the curves via interpolation (see Appendix A.1) to give an age estimate. The median and first standard deviation is found from this via percentiles.

To include: - hot-start - maybe discuss possible solutions and their interpretation

APPENDIX A

INTERPOLATION

The content of Appendix A begins here. Use the `\chapter` command to insert additional appendices.

A.1. Section Name of Appendix A

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